

```

def solve_knapsack():

    val=[50,100,150,200] #value array

    wt=[8,16,32,40] # Weight array

    W=64

    n=len(val) - 1

    def knapsack(W,n): # (Remaining Weight, Number of items checked)

        #base case

        if n<0 or W<=0:

            return 0

        #Higher weight than available

        if wt[n]>W:

            return knapsack(W, n-1)

        else:

            return max(val[n] + knapsack(W-wt[n],n-1),knapsack(W,n-1))

            # max(including , not including)

    print(knapsack(W,n))

if __name__=="__main__":

    solve_knapsack()

```

The screenshot shows a Python Online Compiler interface. The code from the previous block is pasted into the editor. The output panel on the right displays the result of the program execution, which is 350.

```

Python Online Compiler
1 def solve_knapsack():
2     val=[50,100,150,200] #value array
3     wt=[8,16,32,40] # Weight array
4     W=64
5     n=len(val) - 1
6     def knapsack(W,n): # (Remaining Weight, Number of items checked)
7         #base case
8         if n<0 or W<=0:
9             return 0
10
11         #Higher weight than available
12         if wt[n]>W:
13             return knapsack(W, n-1)
14
15         else:
16             return max(val[n] + knapsack(W-wt[n],n-1),knapsack(W,n-1))
17             # max(including , not including)
18     print(knapsack(W,n))
19
20 if __name__=="__main__":
21     solve_knapsack()

```

350